Experiment - 8.1.1. Multiplication ofi Two Square Matrices

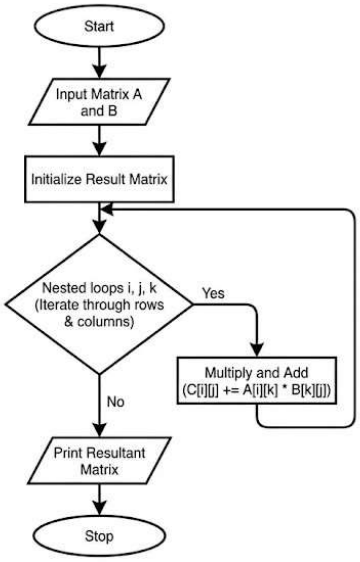
1. Aim

To design and implement a Python program that performs the multiplication of two square matrices of dimension $n \times n$. The program takes the dimension and the elements of both matrices as input from the user and computes their product.

2. Pseudocode

1. **START**
2. **PRINT** "dimension: " and **READ** an integer from the user, storing it in the variable n.
3. **PRINT** "first matrix:" and **INITIALIZE** an empty list A.
4. **FOR** each row i from 0 to n-1:
    - **READ** space-separated integers from the user, convert them into a list, and append to A.
5. **PRINT** "second matrix:" and **INITIALIZE** an empty list B.
6. **FOR** each row i from 0 to n-1:
    - **READ** space-separated integers from the user, convert them into a list, and append to B.
7. **INITIALIZE** a result matrix of size $n \times n$ filled with zeros.
8. **FOR** i in range n (iterating through rows of A):
    - **FOR** j in range n (iterating through columns of B):
        - **FOR** k in range n (iterating through rows of B):
            - **ADD** the product of A[i][k] and B[k][j] to result[i][j].
9. **PRINT** "Resultant Matrix:".
10. **FOR** each row r in the result matrix:
    - **PRINT** the elements of r separated by spaces.
11. **END**

3. Flowchart



4. Python Program

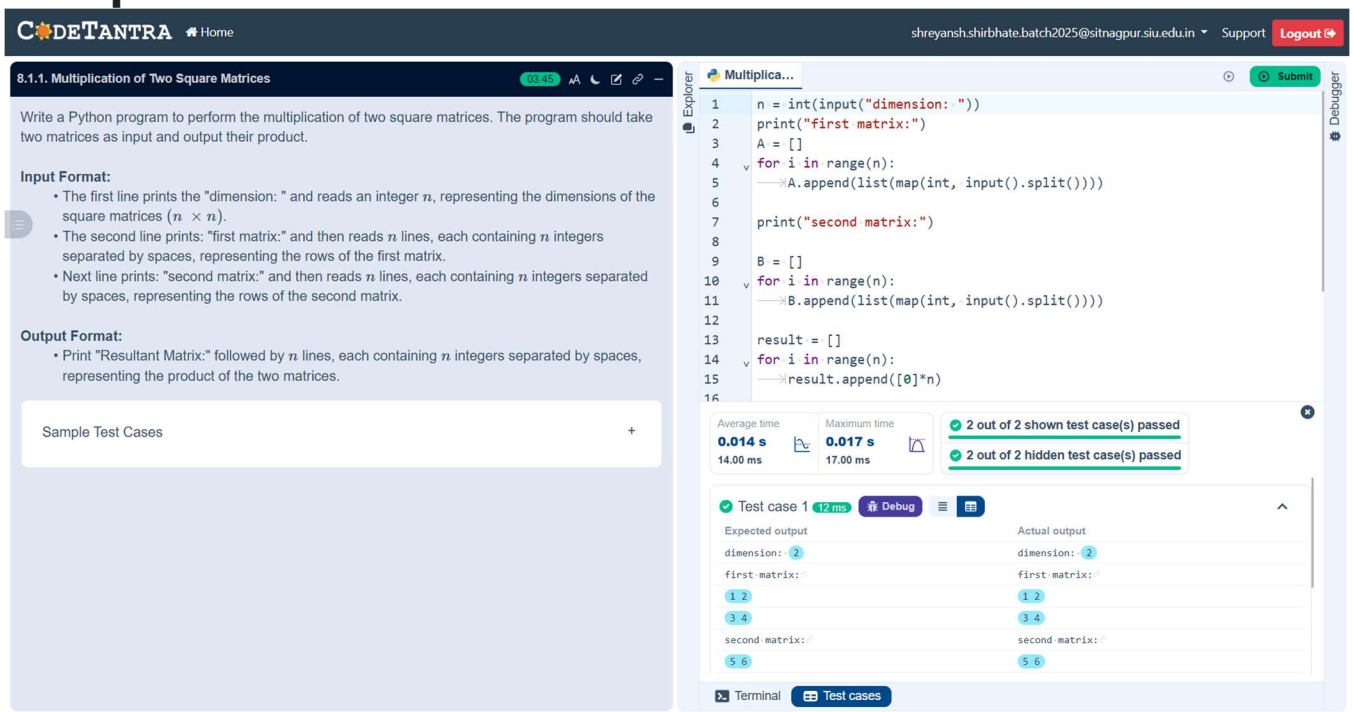
```
n = int(input("dimension: "))

print("first matrix:")
A = []
for i in range(n):
   A. append(list(map(int, input().split())))

print("second matrix:")
B = []
for i in range(n):
   B. append(list(map(int, input().split())))

result = [[0]*n for _ in range(n)]

# Matrix multiplication logic
for i in range(n):
   for j in range(n):
      for k in range(n):
         result[i][j] += A[i][k] * B[k][j]

print("Resultant Matrix:")
for r in result:
   print(*(r))
```

5. Experiment Screenshot